Factors in Economic Growth in Canada

Author(s): Penelope Hartland

Source: *The Journal of Economic History*, Mar., 1955, Vol. 15, No. 1 (Mar., 1955), pp. 13-22

Published by: Cambridge University Press on behalf of the Economic History Association

Stable URL: https://www.jstor.org/stable/2115347

*Factors in Economic Growth in Canada**

WITHIN the sweep of Canada's economic progress, I should like to devote special attention to one period—from about 1900 to the outbreak of World War I—as the period of most rapid growth in the history of Canada since the union of the separate provinces under Confederation in 1867, with the possible exception of the war-stimulated 1940's. I propose to point out certain parallels between that remarkable expansionary period and a much earlier one. I shall also attempt to isolate the important factors responsible for the upsurge of growth after 1900.

To find a previous period of Canadian economic history comparable in the tempo of activity to that of 1900–1914, one must go back to the days before Confederation. The boom of the 1850's was, like that of half a century later, fostered by a large influx of foreign capital and migrants.[1] By contrast, the long period from Confederation to 1900 was characterized by more moderate rates of growth and capital inflow, and an actual net outward flow of migrants. It is instructive to consider the way in which each of the three periods helped to shape the subsequent course of development.

The prosperity of the 1850's was built around railroads and owed much to government facilitation of foreign investment in Canada. The Guaranty Act of 1849 provided a provincial guarantee of interest on the bonds of railroads more than 75 miles long. By 1851, the demand for this assistance had become so great that the privilege was thereafter restricted to trunk lines, but the Canadian Government authorized municipalities to take the securities of secondary roads that had received a government charter to build feeder lines. A Consolidated Municipal Loan Fund was organized to help the cities borrow in London. Prosperity in England and the British investing public's boundless enthusiasm for railroad shares made the Canadian securities easily marketable. The result was a tremendous volume of railroad building and other capital investment. Whereas in 1850 only about 60 miles of rail-

* This article is a revision of some comments made at the meeting of the Committee on Economic Growth of the Social Science Research Council, New York, April 9–10, 1954.

[1] Up to World War II, there were four decades in which Canada had net inward migration: 1851–61, 1901–11, 1911–21, and 1921–31. D. C. Corbett, "Immigration and Economic Development," *Canadian Journal of Economics and Political Science* (hereafter referred to as *CJEPS*), XVII (August 1951), 360 ff. In the last two of these decades, however, the rate of net immigration was lower, and it was not accompanied by any sizable net capital inflow.

road had been in operation in Canada, by 1860, 2,000 miles had been added; and, in the estimate of the Minister of Finance, over $100 million had been spent on their construction. It has also been estimated that as much foreign capital was invested in Canada per capita during the 1850's as in the decade from 1904 to 1914.[2]

The demand for labor and materials of course skyrocketed. The extension of railroads into new areas stimulated the growth of cities, and precipitated a rolling real-estate boom with land values inflated by speculation. Rising land values meant higher tax assessments and thus increasing ability for the municipalities to borrow in London. According to the Mayor of Hamilton at the time, interest on previous borrowing was always punctually paid out of the proceeds of the latest loan.

When domestic inflation brought rising costs, and the end of the Crimean War caused a drop in the price of wheat and made further loans in London difficult, the crisis of 1857–1858 occurred. The banks, which had invested heavily in real-estate mortgages, curtailed credit sharply, causing a collapse of land values, and Canada for the first time in its history experienced bank failures. All the railroads were in trouble, and the municipalities were in a dilemma. If they raised taxes sufficiently to meet their interest payments, they would force people out of the cities and cause land values to fall further; but if they did not, they could not meet their obligations. The cities decided that the English bondholders could wait. Among the railroads, only two short lines had paid any dividends and it was extremely doubtful that a number would be able to meet their interest payments. This financial collapse had a serious and lasting effect on Canadian credit. The provincial government had to aid both railroads and municipalities, but with depressing effects on its own credit. In 1866, for example, it was unable to raise more than half of a medium-size loan, even while offering 8 per cent.

The period from 1867 to 1900, which witnessed sustained growth of the Canadian economy, can be considered an era during which the gains so tumultuously harvested during the 1850's were consolidated and extended at a more moderate pace. The real capital formation of the 1850's, however, provided the foundation for the subsequent increases in output and productivity.

[2] A. Shortt, "Railroad Construction and National Prosperity," in *Proceedings of Royal Society of Canada,* December 1914; see also A. Shortt, "History of Canadian Currency, Banking and Exchange," *Journal of the Canadian Bankers' Association,* January 1904, pp. 106 ff. and April 1904, pp. 199 ff.; H. C. Pentland, "Role of Capital in Canadian Economic Development before 1875," *CJEPS,* XVI (November 1950), pp. 457–74.

The entire period from Confederation to the turn of the century saw continuous growth in the Canadian economy. Real consumption per capita increased at an average annual rate of nearly 3 per cent in that period, as compared with about 1.5 per cent in 1900–1920 and 2½ per cent for the period 1920–1952.[3] Rates of growth in general appear much higher in the 1870's than in either the 1880's or the 1890's. For example, real output per capita increased at an average rate of over 2½ per cent a year in the 1870's, 1½ per cent in the 1880's, and a little over 2 per cent in the 1890's. Total real national product grew during the 1870's at the very respectable annual average rate of about 5 per cent.

It should be noted, however, that the available data probably overstate the over-all growth of output in the period from Confederation to 1900. The Canadian economy was rapidly changing from one largely composed of self-sufficient households to a market economy, which fact implies that the coverage of the data was becoming more complete as time went on. In addition, the differences in the rates of growth during each of the three decades probably are influenced by cyclical movements, in distinction to any shift in underlying economic trends.[4]

While the growth in the last third of the nineteenth century occurred in the face of net emigration from Canada, the expansion of the first 14 years of the twentieth century was supported by a huge influx of immigrants. Indeed, the chief difference between the two periods, 1867–1900 and 1900–1914, seems to lie in the degree of foreign stimulus provided to the Canadian economy. The post-1900 expansion was based on a huge influx of both foreign capital and labor, which provided the basic productive factors necessary for the cultivation of the western prairies and the accompanying development of other branches of the Canadian economy. Growth in the three decades before 1900 had been based far more on Canadian resources, although even in that period investment by foreigners (under the stimulus of government borrow-

[3] These and the following data on average rates of increase per annum are derived from O. J. Firestone, *Canada's Economic Development 1867–1952,* a study prepared for the Third Conference of the International Association for Research in Income and Wealth, September 1953, possibly to be published in 1955.

[4] The evidence lies largely in the fact that the volume of capital inflow from abroad in 1900 was considerably smaller, and had been declining for some years below the level of the late 1880's and early 1890's. In 1880 and again in 1900, Canadian payments of interest and dividends abroad considerably exceeded the net capital inflow, while the opposite was true in 1890. Both 1870 and 1880 mark troughs in the cycles of net capital inflow. P. Hartland, *The Canadian Balance of Payments Since 1868,* a study to be published shortly by the National Bureau of Economic Research, Inc.

ing) played an important part in real capital formation.[5] According to the usual measures of over-all economic growth, Canada appears to have grown faster between 1900 and the outbreak of World War I than in any other period since the Act of Confederation in 1867. This statement can be substantiated in terms of the rates of increase of real national product, population, labor force, occupied farm acreage, and such. Moreover, in this period the rate of net immigration was at its peak and the inflow of foreign capital into Canada was larger, in proportion to national product, than at any other time since Confederation.[6]

Real national product grew at an annual average rate of nearly 8 per cent in the first decade of this century; population at more than 3 per cent; real output per capita at 3½ per cent; real output of manufacturing at 7 per cent (so far as the data permit an estimate); and the labor force at more than 5 per cent. In the five years from 1906 to 1910, the inflow of foreign capital averaged 9 per cent of gross national product and in the next five years 12 per cent.

These are impressive figures.

In view of the sustained growth that had preceded the post-1900 expansion, one cannot avoid the question, "Why did not the boom occur earlier?" The question becomes even more compelling when one recalls the social milieu of the late nineteenth century and the goal of Canadian policy so vigorously pursued in those years. The mobility of European peoples during the nineteenth century is well known. Large waves, not only of labor, but also of capital, left Europe, and especially Great Britain, throughout the century for other underdeveloped countries like Canada. Especially after 1870, both Australia and Argentina (countries like Canada, possessed of rich natural resources but acutely lacking in labor and capital), experienced a decade of rapid expansion marked by a great influx of people and foreign capital, the opening of new lands for the production of food and materials which found a profitable market in Europe, and a rapid increase of total output. In these cases (as in Canada during the 1850's) the boom period was marked by inflationary price rises, especially of real-estate values, and a final financial collapse which occurred when interest payments on for-

[5] For a discussion of the role of the Canadian Government in development, see W. T. Easterbrook, "State Control and Free Enterprise," in *Progress of Underdeveloped Areas,* ed. B. F. Hoselitz (Chicago: University of Chicago Press, 1952); P. Hartland, "Private Enterprise and International Capital," *CJEPS,* February 1953, pp. 70–80.

[6] The huge inflow in 1950 of about $1 billion was only about 5 per cent of gross national product.

eign indebtedness became greater than the proceeds of the new borrowings. These waves of people and capital leaving Europe before 1900 did not settle in Canada in any sizable numbers, despite the fact that Canada was eager to attract foreign capital and immigrants, and made strenuous efforts to that end. The Act of Confederation in 1867, for example, was in large part a tool to aid in economic development. In fact the late Professor Innis described it as a means of mobilizing the credit of the provinces for the pursuit of developmental policies.[7]

Confederation involved the new Dominion Government in commitments for an extension of the existing railway net, to join different sections of the country in order to expand Canadian markets. The emphasis placed on railroad building, and the avid zeal of Canadian leaders for economic expansion throughout this period, were based in the desire to emulate the United States and to capture part of the fast-growing United States Market. As one writer put it, "Had there been no American railroads, there would still have been a desire for railroads in Canada, but no such pressing need for them."[8]

The so-called "National Policy" established in 1879, which provided tariff protection and government subsidies to encourage domestic industries, government propaganda campaigns carried on in the United States and Europe to recruit immigrants, the continuation of the Parliamentary Act of 1849 providing a government guarantee of interest on railroad bonds, the development of various forms of state aid to farmers—these are but some of the high lights of a government policy aimed directly at the further development of the country.

The surprising thing is that despite all these efforts and despite the mobility of European capital and labor, Canada did not succeed, before 1900, in attracting foreign immigrants or capital on anything like the scale that might have been expected. Why was this?

Undoubtedly part of the slow response is to be explained in terms of a healthy caution on the part of British investors burned in the late 1850's. But probably caution was a minor factor. To be sure, the Grand Trunk Railroad had been widely heralded in England as a sure road to riches—in fact an 11½ per cent return was promised—and English interests strongly resented the fact that the Canadian Government had forced a reorganization of the road's capital structure involving lower

[7] H. A. Innis, *Problems of Staple Production* (Toronto: Ryerson Press, 1933).

[8] G. P. de T. Glazebrook, *History of Transportation in Canada* (Toronto: Ryerson Press; New Haven: Yale University Press for the Carnegie Endowment for International Peace, 1938), p. 161.

interest rates and the exchange of bonds for stock. In 1885 upon the completion of the main lines of the Canadian Pacific Railroad ahead of schedule, the *Economist* wrote, "It is impossible to refer in any favorable way to the commercial prospects of the Canadian Pacific." [9] But English investors were an optimistic crew, incapable of resisting railroad securities, and although at first the Canadian Pacific had difficulty in marketing its shares in London, as early as 1883 17 per cent of its stockholders were located in England.[10]

For the period 1868 through 1899 Canadian receipts from the flotation of new railroad securities in London did represent a discount of about 17 per cent on their face value. Other issues publicly marketed abroad in these years—they were primarily government bonds—were sold nearly at par and a total of over $800 million of new issues was marketed in England.[11]

One also has the feeling that too much stress has been placed on the importance of the closing of the American frontier in the timing of the Canadian expansion. While the American West was still being occupied and cultivation extended, the European market for food, especially wheat, was expanding rapidly enough to absorb the huge increases in output that resulted from the opening and populating of Argentina and Australia. In the decade of the 1880's, Australian production of wheat quadrupled,[12] and during the 1890's Argentine wheat exports more than tripled, from just over 300,000 tons in 1891 to 1,900,000 tons in 1900.[13] Meanwhile Canadian wheat exports remained almost unchanged. Thus, if other factors had been propitious, the fact that there was still uncultivated free land in the American West would not have deterred the opening of the Canadian prairies in the late 1880's or 1890's.

But the other factors were *not* propitious. Although technically the prairie provinces were "opened" in 1885 upon the completion of the Canadian Pacific Railroad, little land was taken up by homesteaders, until the turn of the century.

The hardships of life on the frontier, breaking the prairie soil and preparing the land, and the disappointments of the farmers when the methods of cultivation they had used in the East or in Europe failed

9 XLIII (May 23, 1885), 625.

10 As compared with 63 per cent in 1915. H. A. Innis, *History of the Canadian Pacific Railroad* (Toronto: McClelland and Stewart Ltd., 1923), p. 276.

11 Hartland, *Canadian Balance of Payments Since 1868.*

12 G. Fitzpatrick, *British Empire in Australia, 1834–1939* (Melbourne: Melbourne University Press, 1941), pp. 137, 171.

13 J. H. Williams, *Argentine International Trade Under Inconvertible Paper Money, 1880–1900* (Cambridge: Harvard University Press, 1920).

to yield an adequate harvest, caused many to give up and leave, often apparently for the United States. Their disillusionment was communicated to their friends and relatives in Europe.[14]

The most important determinant of the Canadian expansion after 1900 seems to lie in technological improvements in methods of cultivation which finally permitted a profitable harvest from the prairie provinces. If summer fallow and dry farming techniques had been used extensively earlier, if Marquis wheat had been adopted or developed sooner, it seems probable that yields would have been sufficiently sizable and quality sufficiently good to compensate for the higher railroad and transportation rates that prevailed in the 1890's. The roller method of flour milling, which made spring wheat better suited to grinding than winter wheat, was perfected in 1881, and the chilled steel plow capable of turning the prairie sod was available in 1870. Without these other techniques of cultivation, however, the distance of the prairies from open water transport and the fact of relatively high Canadian railroad rates in the 1890's delayed the opening of the West. When, however, the Crow's Nest Pass Agreement (1897) brought lower railroad rates on wheat for export and when the general use of steel hulls and steam turbines brought a decline in ocean freights and insurance, the effective supply of economically cultivatable Canadian wheat land was greatly augmented. Word of the profits to be obtained from growing wheat in the Canadian prairie provinces spread rapidly and the labor needed to do it came.

Not only in the field of wheat culture did technological advance stimulate the capital investment in Canada in these years, but also in mining and metallurgy. A predecessor of the International Nickel Company, for example, originally sought to develop certain copper deposits in Canada. Their mines, however, were not profitable until methods for separating copper from the accompanying nickel were devised, and then their interest was shifted to the by-product, nickel, via an innovation in armament production which the United States Government adopted. After 1890 the tremendous progress in the United States in types of mining machinery was basic to American interest in and investment in Canadian mines in the twentieth century.

World War I shut off the inflow of capital and labor, but—in con-

[14] On the other hand, one could say that if the United States frontier had been closed at this time there might have been more sustained attempts at wheat growing, and perhaps the innovations which later made wheat culture profitable would have occurred earlier, being mothered by necessity.

trast to 1857—without a financial crisis and while the average capital inflow was still considerably in excess of interest and dividend payments due abroad. From 1910 through 1913 the capital inflow averaged nearly $400,000,000 a year, and interest and dividend payments only about $115,000,000.[15] By the war's end Canada had become primarily a capital-exporting country, with maturities of past issues and purchases of foreign securities exceeding new foreign investment in Canada except in isolated years.

From the end of the war to the depression of the 1930's, Canada again experienced a period of consolidation and extension of previous gains analogous to that ensuing after the boom of the 1850's. The larger domestic market brought an expansion of manufacturing activity in the East, but the purely Canadian growth in these years was almost entirely based on wheat. The growth of the automobile, paper, and mineral industries, all of which gave diversity to the economy, was primarily supported by United States demand and United States investment.

There is a temptation in this field of development to seek for generalizations which, in a pat fashion, explain the whole case. Yielding to this imp, may I proffer the tentative hypothesis that the best part of Canadian economic growth is to be explained in terms of its technological "Southern exposure"—easy entry to American innovation. Given the fact that Canada is richly equipped with natural resources, Canadian output has grown at a rapid long-run rate because American innovation has made Canadian resources economically productive, and often because Americans were willing to risk capital in order to develop these resources.

The United States, for example, is generally recognized as the leader in the marvelous development of mining machinery that has occurred since 1890. Although Canada produces most of the machines and supplies used in its mining and milling, the United States has been directly or indirectly responsible for their development. They are manufactured in Canada by subsidiaries of American companies, or by Canadian companies that have procured the right to manufacture.[16] Or again, the shift in Ontario from wheat to dairy products, especially cheese, that was partly responsible for the increases in output and income in the 1880's was not successful until improved factory devices developed

[15] Hartland, *Canadian Balance of Payments Since 1868.*

[16] E. S. Moore, *American Influence in Canadian Mining* (Toronto: Ryerson Press; New Haven: Yale University Press for the Carnegie Endowment for International Peace, 1941), p. 90.

in New York were adopted.[17] The techniques of summer fallow and dry farming, necessary to successful wheat culture in the prairies, had been used by the Mormons in Utah for some years before they were adopted in Canada, and Red Rife seed itself successfully used at the turn of the century, although originally received in Ontario from Scotland, was developed in the United States and imported into Manitoba from there.[18] American agricultural machinery has always been popular in Canada. During the years 1926–1933 the International Harvester Company of Canada had 40 per cent of the total domestic agricultural implement business there, although it did have strong competition, especially in foreign markets, from Massey-Harris, whose total sales were 50 per cent greater.[19] These examples could be multiplied.

But all this raises more questions than it answers. Were these decades of inflationary boom and tremendously rapid expansion sound in the sense that the growth which they brought represented an adequate distribution of productive resources in terms of long-run rates of return?

What was the role of government policy in bringing about the period of expansion? Not only in the two Canadian cases, but in the case of Australia and Argentina, too, the inflow of foreign capital to which the boom was tied in preponderant part took the form of government borrowing abroad both in the flotation of its own issues and in its guarantee of other securities. Yet we have seen in Canada during the period 1870–1900 that government borrowing abroad and government encouragement of development were not sufficient to attract the necessary immigration.[20] On balance, was the protective tariff an aid or an impediment? In Canada this has long been a controversial question.

What was the role of English investment houses in those periods of economic expansion that occurred in the nineteenth century? The English investing public did seem possessed of a band-wagon psychology in regard to foreign issues; the English banking houses did not seem especially eclectic in some of the foreign issues they brought out.

[17] O. D. Skelton, *General Economic History of the Dominion, 1867–1912* (Toronto: Publisher's Association of Canada, 1913), p. 119.

[18] G. E. Britnell, *The Wheat Economy* (Toronto: University of Toronto Press, 1939), pp. 35 ff.; A. H. R. Buller, *Essays on Wheat* (New York: Macmillan Co., 1919), pp. 206, 215; L. D. Nesbitt, *The Story of Wheat* (Calgary: Alberta Wheat Pool, 1946), p. 11.

[19] H. Marshall, F. A. Southard, and K. W. Taylor, *Canadian American Industry* (Toronto: Ryerson Press; New Haven: Yale University Press, 1936), p. 61.

[20] In this regard, however, it should be pointed out that although the government's campaign overseas to encourage the migration of people to Canada continued throughout, its activities were more aggressive in the early 1880's and after 1897, but in the years between were rather stereotyped, according to one source. Skelton, *General Economic History of the Dominion*, p. 193.

And what is the explanation for the very sudden shift from a very sizable debit balance of payments on current account before World War I to a moderately sizable credit balance, not only during the war, but also in the years since?

How has the operation of the Canadian economy been influenced by developments in the United States? In this regard one has the feeling that a complete understanding of Canadian economic development is dependent as much on an analysis of the differences between Canadian and United States cultures as on an analysis of their similarities.

PENELOPE HARTLAND, *Washington, D.C.*